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1. A researcher is modeling the effect of two predictors (e.g., hours studied and number of practice exams) on student test scores using Bayesian linear regression: 3 / 3 points

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

The researcher sets:

• **Normal priors** on $\beta_0, \beta_1, \beta_2$, i.e., $\beta_j \sim \mathcal{N}(0, 10^2)$.
• An **inverse-gamma prior** on σ^2 , i.e., $\sigma^2 \sim \text{Inv-Gamma}(2, 2)$.

If the researcher wants to predict a new student's test score with uncertainty, which approach should they take?

- ☒ Sample values of $\beta_0, \beta_1, \beta_2$ from the posterior and compute the predictive distribution of Y_{new} .
☐ Use the posterior mean of $\beta_0, \beta_1, \beta_2$ to compute the point estimate.
☐ Ignore the Bayesian framework and compute a frequentist confidence interval.
☐ Use the MAP estimates to compute a single best prediction.

☒ Nice work
Nice!

2. **True or False:** Consider the normal data model with unknown mean μ and unknown variance σ^2 . In a setting where only σ^2 is of primary interest, μ is considered a nuisance parameter. 3 / 3 points

- ☒ True
☐ False

☒ Nice work
Nice!

3. Consider the normal data model with unknown mean μ and unknown variance σ^2 . To find the marginal posterior of σ^2 , μ is considered a nuisance parameter. 3 / 3 points

- ☐ Take the derivative of the joint posterior of μ and σ^2 with respect to μ .
☐ Take the derivative of the joint posterior of μ and σ^2 with respect to σ^2 .
☒ Integrate the joint posterior of μ and σ^2 with respect to μ .
☐ Integrate the joint posterior of μ and σ^2 with respect to σ^2 .
☐ None of the above.

☒ Nice work
Nice!

4. A researcher is modeling the effect of two predictors (e.g., hours studied and number of practice exams) on student test scores using Bayesian linear regression: 3 / 3 points

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

The researcher sets:

• **Normal priors** on $\beta_0, \beta_1, \beta_2$, i.e., $\beta_j \sim \mathcal{N}(0, 10^2)$.
• An **inverse-gamma prior** on σ^2 , i.e., $\sigma^2 \sim \text{Inv-Gamma}(2, 2)$.

True or False: The posterior distribution of σ^2 depends on the observed values of X_{i1} and X_{i2} .

- ☒ True
☐ False

☒ Nice work
Nice!

5. In Bayesian inference for a normal model with unknown variance σ^2 , the inverse chi-squared distribution is often used as a prior. What is a key reason for choosing this prior? 3 / 3 points

- ☒ It is the conjugate prior, making the posterior distribution also inverse chi-squared.
☐ It is equivalent to a normal distribution, making it easier to interpret.
☐ It simplifies calculations because the posterior mean is always equal to the prior mean.
☐ It ensures that the posterior of σ^2 remains independent of the observed data.

☒ Nice work
Nice!

6. A researcher is modeling the effect of two predictors (e.g., hours studied and number of practice exams) on student test scores using Bayesian linear regression: 3 / 3 points

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

The researcher sets:

• **Normal priors** on $\beta_0, \beta_1, \beta_2$, i.e., $\beta_j \sim \mathcal{N}(0, 10^2)$.
• An **inverse-gamma prior** on σ^2 , i.e., $\sigma^2 \sim \text{Inv-Gamma}(2, 2)$.

True or False: The marginal posterior distribution of β_0 given the data will be independent of β_1 .

- ☐ True
☒ False

☒ Nice work
Nice!

7. **True or False:** The choice of prior for one parameter does not affect the posterior distribution of other parameters in a Bayesian model. 3 / 3 points

- ☐ True
☒ False

☒ Nice work
Nice!

8. Suppose we have a Bayesian model with two parameters θ_1 and θ_2 , and their joint posterior is given by $\pi(\theta_1, \theta_2 | y)$. To obtain the marginal posterior of θ_1 , we: 3 / 3 points

- ☐ Use the prior $\pi(\theta_1)$ alone.
☒ Compute $\pi(\theta_1 | y) = \int \pi(\theta_1, \theta_2 | y) d\theta_2$.
☐ Find the maximum of the joint posterior over θ_2 .
☐ Assume θ_2 is independent and ignore it.

☒ Nice work
Nice!

9. Consider the normal data model with unknown mean μ and unknown variance σ^2 . To simulate draws from a joint posterior distribution: 3 / 3 points

- ☐ First simulate a value from the prior distribution on σ^2 , and then simulate a value from the distribution of μ conditional on σ^2 .
☒ First simulate a value from the marginal posterior distribution on σ^2 , and then simulate a value from the distribution of μ conditional on σ^2 .
☐ First simulate a value from the prior distribution on μ , and then simulate a value from the distribution of σ^2 conditional on μ .
☐ None of the above.

☒ Nice work

10. **True or False:** In Bayesian inference for a normal model 3 / 3 points

$$Y_i | \mu, \sigma^2 \sim \mathcal{N}(\mu, \sigma^2)$$

If the prior for σ^2 follows an inverse chi-squared distribution, then the posterior distribution of σ^2 given the data follows a chi-squared distribution, regardless of whether μ is known or unknown.

- ☐ True
☒ False

☒ Nice work
Nice!